MUTUAL FUND AND STOCK RISK ANALYTICS USING HADOOP AND SPARK

A PROJECT REPORT

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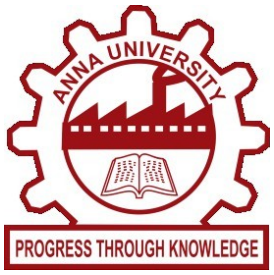
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ABSTRACT

Investors usually choose mutual funds and stocks by looking only at past returns, without seeing the risk behind them. This project builds a big data platform that analyses Indian mutual funds and Nifty 50 stocks to show both return and risk. Twenty years of daily NAV data published by AMFI (38,279 schemes and about 37 million records) and ten years of Nifty 50 share prices and company balance sheets are collected using Python and stored in the Hadoop Distributed File System (HDFS). Apache Spark cleans the data and organises it into Bronze, Silver and Gold layers stored as Parquet files. Spark then calculates returns (CAGR), risk measures such as volatility, maximum drawdown, Value at Risk and beta, a 0–100 risk score, holding-period returns and company debt ratios. Machine learning models from Spark MLlib (Linear Regression, Random Forest, Gradient-Boosted Trees and Logistic Regression) and ARIMA are compared using walk-forward validation, and Random Forest gave 5.9% lower error than a simple baseline. The results are served through a FastAPI service protected by an API key and shown on a website and a Power BI dashboard, helping investors make better long-term decisions.

Keywords

Hadoop HDFS, Apache Spark, Mutual Fund Risk Analysis, Spark MLlib, Medallion Architecture, Power BI

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CHAPTER 1

INTRODUCTION

The Indian stock market and the mutual fund industry have grown rapidly in the last decade, and millions of new investors now put their savings into mutual funds and shares. Most of them still choose an investment by looking at last year's return, without understanding how risky it is or how long they should stay invested. At the same time, the data needed to answer these questions is huge: AMFI publishes a new Net Asset Value (NAV) for every scheme on every working day, and stock exchanges publish daily prices for thousands of companies. This project uses big data tools to turn this raw data into clear return, risk and prediction results that an ordinary investor can understand.

1.1 Background

A mutual fund's NAV is the price of one unit of the fund. More than 38,000 schemes have published NAVs since 2006, giving over 37 million daily records. Stocks of the Nifty 50 companies add ten years of daily prices, and each company also publishes yearly balance sheets that show how much debt it carries. Traditional tools such as spreadsheets cannot open data of this size; Excel, for example, stops at 1,048,576 rows. Big data technologies such as the Hadoop Distributed File System (HDFS) and Apache Spark were designed to store and process data at this scale by splitting the work across many machines, and the same programs run on a single laptop or on a large cluster.

1.2 Motivation

Many investors exit a fund after one bad year and lose the long-term benefit of staying invested. Others pick a fund with a high return without knowing that it fell by more than a third during the 2020 crash. We wanted to show, with real data, how risky each fund and stock has been, whether holding for longer really pays,

and whether machine learning can predict returns better than a simple guess. Building this as a complete pipeline also gave us hands-on experience with the big data tools taught in our Big Data Architecture course.

1.3 Objectives

- To collect 20 years of mutual fund NAV data and 10 years of Nifty 50 stock and company debt data and store it in Hadoop HDFS.
- To clean and process the data with Apache Spark using a Bronze, Silver and Gold (Medallion) data lake design.
- To calculate returns, risk measures and a 0–100 risk score for every fund and stock.
- To measure how often holding an investment for 1, 3, 5, 7 and 10 years made money.
- To compare forecasting models fairly against a baseline and to predict each fund's risk band for the next quarter.
- To serve the results securely through an API with a key and show them on a website and a Power BI dashboard.

1.4 Problem Statement

Investors do not have a single place where the return and the risk of a mutual fund or stock are shown side by side. Raw NAV and price data is too large for ordinary tools, contains duplicate and incorrect values, and is spread across different sources and formats. Claims such as "holding for the long term always pays" are rarely backed by evidence, and prediction models are often published without checking whether they beat a simple guess. There is a need for a scalable system that stores and processes this data, produces explainable risk scores and tested predictions, and presents them securely in an easy-to-understand dashboard.

1.5 Scope of the Project

The project covers all Indian mutual fund schemes published by AMFI, with detailed analysis of 1,137 growth funds that have a full ten-year history and a curated set of 30 popular funds (large cap, flexi cap, mid cap and index funds). It also covers the 50 companies of the Nifty 50 index, the Nifty 50 and Sensex indices, and about four years of company debt data. The system runs on a single-node Hadoop cluster inside WSL Ubuntu, and the results are available on a local website and in Power BI. The design can be extended to a multi-node cluster, daily scheduled updates and public hosting without changing the processing logic.

CHAPTER 2

LITERATURE SURVEY

Research on stock market and mutual fund analytics has moved from manual ratio analysis to machine learning and big data platforms. To understand recent progress and gaps, we studied five IEEE papers published between 2024 and 2026. They cover mutual fund risk analysis, stock price prediction using artificial intelligence, Spark-based stock forecasting, big data risk assessment in financial markets, and real-time forecasting frameworks.

Dasgupta et al. [1] studied investment risk in the Indian mutual fund industry using machine learning. They compared AdaBoost, Support Vector Machines, Gradient Boosting and Random Forest models and evaluated them with MSE, RMSE, MAE and MAPE. Random Forest performed best, with an R^2 of 0.925 and a MAPE of 0.096. This result supports our choice of Random Forest as one of the main models. Our project extends this idea by computing risk for over a thousand funds with Apache Spark and by testing models year by year on data they have not seen.

Al-Alawi and Alshakhoori [2] carried out a systematic literature review of stock price prediction using artificial intelligence. They selected fourteen influential papers from 2020 to 2023 covering technical, fundamental and sentiment-based approaches. They found that hybrid models using deep learning handle the non-linear and time-dependent behaviour of markets better than traditional methods, and they pointed to integrating big data analytics as a promising direction. This motivated us to build our analysis on a big data pipeline and to compare several models instead of relying on one.

Koduppanapolackal et al. [3] proposed a big data method for stock price prediction in which historical stock data is processed with Apache Spark and combined with real-time data from the Alpha Vantage API. A Long Short-Term

Memory (LSTM) network was trained and evaluated using MSE and RMSE, and the predictions were visualised with Matplotlib. Their work shows that Spark is suitable for preparing large financial datasets. Our project uses Spark in the same way, and adds HDFS storage in Bronze, Silver and Gold layers, risk metrics and a secured dashboard.

Arumugam, Balasingam and Agila [4] presented big data frameworks for advanced risk assessment and fraud detection in financial markets using Recurrent Neural Networks (RNN). Their work highlights that financial risk must be assessed over very large volumes of market data, which requires big data processing rather than traditional tools. Our project also focuses on risk, but uses an explainable risk score built from volatility, maximum drawdown, Value at Risk, beta and company debt, so that each score can be explained to an investor.

Udayakumar and Anbalagan [5] designed an AI-driven real-time stock forecasting framework that uses Apache Kafka for data ingestion, Cassandra for storage and Spark streaming for processing. Forecasts come from an ensemble of LSTM networks and Random Forests. Tested on S&P 500 tick data from 2015 to 2024, the system reported latency below 50 milliseconds, a throughput above 2.5 million events per second, and robustness during the 2020 crash and the 2022 interest rate rise. This shows that Spark and Random Forest scale to very large market data. Our project applies similar tools to daily Indian fund and stock data in batch mode, which suits long-term investors at a much lower cost.

Research Gap

Most of these studies focus on the accuracy of price prediction for stocks. Few of them combine the full history of Indian mutual funds, an explainable risk score, evidence on holding periods, company debt analysis and an honest comparison of models against a simple baseline in one system. Results are also rarely served securely to business intelligence tools. Our project fills this gap with a complete

Hadoop and Spark pipeline whose results are delivered through a key-protected API to a website and a Power BI dashboard.

CHAPTER 3

ARCHITECTURE DIAGRAM

The system is designed as a layered big data pipeline. Data flows from the public sources, through ingestion, into the Hadoop data lake, where Apache Spark refines it layer by layer. The final results are served through an API to the website and Power BI. Figure 3.1 shows the complete architecture.

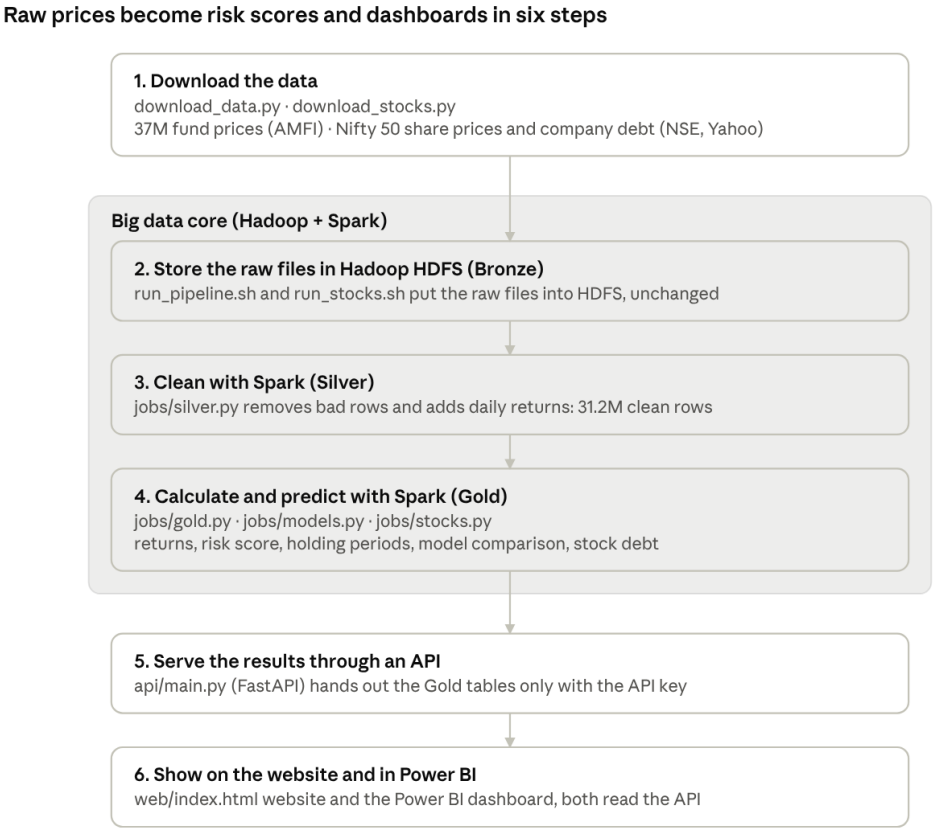


Figure 3.1 Architecture diagram

3.1 Data Source and Ingestion Layer

Python scripts download the AMFI NAV archive for every mutual fund since 2006, the official Nifty 50 company list from NSE, and daily stock prices, index values and yearly balance sheets from Yahoo Finance. The downloads can resume if interrupted, and every file is logged in a manifest.

3.2 Storage Layer (Hadoop HDFS)

All raw files are copied unchanged into HDFS, forming the Bronze layer. HDFS stores large files in blocks with fault tolerance and allows Spark to read them in parallel. The data is kept in Apache Parquet format, which is compressed and column-based, and partitioned by year so that Spark reads only what each query needs.

3.3 Processing Layer (Apache Spark)

Spark jobs clean the data into the Silver layer by removing duplicate and incorrect prices and calculating daily returns. Further jobs compute returns, risk measures, the risk score, holding-period statistics, company debt ratios and machine learning results, and write them as small tables in the Gold layer.

3.4 Serving Layer (FastAPI)

A FastAPI web service reads the Gold tables and makes them available at fifteen data addresses such as /funds, /risk-scores and /stocks. Every request must carry a secret API key in the X-API-Key header; a missing or wrong key is refused with error 401.

3.5 Visualization Layer

The website and the Power BI dashboard both read the same API. The website shows the growth of an investment, top funds, risk charts, holding-period results, model comparison and company debt. Power BI provides four interactive pages: Home, Risk, Holding Period and Models.

CHAPTER 4

SYSTEM ANALYSIS

4.1 Existing System

In the existing approach, investors and many advisors compare funds using return tables published on fund websites and in newspapers. These tables show past returns but rarely show volatility, the worst fall, or how often an investment lost money. Analysis is usually done in spreadsheets, which cannot handle tens of millions of NAV records, so only a small sample of funds or a short period is studied. Company debt is checked manually from annual reports. Prediction models, when used, are often tested on shuffled data, which lets them see the future and makes their accuracy look better than it is. Reports are static, and there is no secure way to share live results with other tools.

4.2 Proposed System

The proposed system stores the complete history of all mutual funds and the Nifty 50 stocks in Hadoop HDFS and processes it with Apache Spark. It produces, for every fund and stock, returns over 1, 3, 5 and 10 years, risk measures, a 0–100 risk score with a Low, Medium or High band, and holding-period evidence. Company debt is judged using debt-to-equity and interest cover. Five forecasting models and three risk-band models are compared with walk-forward validation, always against a simple baseline. All results are served through an API protected by a key and displayed on a website and a Power BI dashboard. The whole pipeline runs with one command and can be repeated whenever new data is published.

4.3 System Requirements

Hardware: a laptop or desktop with at least 8 GB of RAM, 4 GB of free disk space and a multi-core processor, running Windows 10 or 11 with WSL, or any Linux system. An internet connection is needed only for downloading data.

Table 4.1 Software requirements

Software	Purpose
WSL Ubuntu, VS Code	Operating environment and code editor
Java 11, Python 3.12	Runtime for Hadoop, Spark and scripts
Hadoop 3.4.3 (HDFS)	Distributed storage of the data lake
Apache Spark 3.5 (PySpark), Spark MLlib	Data processing and machine learning
pandas, pyarrow, requests, yfinance, statsmodels	Data collection, file handling and ARIMA
FastAPI, Uvicorn	API with key-based security
HTML, CSS, JavaScript, Chart.js	Website
Power BI Desktop	Interactive dashboard

CHAPTER 5

MODULES DESCRIPTION

5.1 Data Collection Module

This module (download_data.py and download_stocks.py) gathers the raw data. It downloads the compressed AMFI NAV archive, decompresses it and converts it into Parquet files partitioned by year, along with the scheme list and the curated list of 30 funds. For stocks, it reads the official Nifty 50 list from NSE and downloads ten years of daily prices for each company and for the Nifty 50 and Sensex indices, plus yearly balance sheet figures such as total debt, equity, operating profit and interest expense. The module retries failed downloads, skips files already downloaded the same day, and records every file in a manifest.

5.2 Data Preprocessing Module

This module (silver.py) is a Spark job that turns the Bronze data into the Silver layer. It removes duplicate rows and invalid NAVs, keeps data from 2013 onwards, and calculates the daily return as today's NAV divided by the previous NAV minus one, using a Spark window for each scheme. A daily change above 50%, or a change after a gap of more than ten days, is treated as a data error and blanked. The output is 31.2 million clean rows for 29,097 schemes, stored in Parquet and partitioned by year.

5.3 Analytics and Risk Scoring Module

This module (gold.py and stocks.py) computes the main results in the Gold layer. For each fund and stock it calculates the 1-year return, the 3, 5 and 10-year compound annual growth rate (CAGR), and the value of ₹1 lakh invested ten years ago. Risk is measured by volatility, maximum drawdown, Value at Risk and Conditional Value at Risk at 95%, beta and alpha against the Nifty 50, and the Sharpe and Sortino ratios. Each measure is ranked against peers and combined

into a 0–100 risk score with weights of 30 (volatility), 25 (drawdown), 15 (VaR) and 15 (beta), plus 15 for debt-to-equity in the case of stocks. The module also takes every month since 2013 as a starting point to measure how often holding for 1, 3, 5, 7 and 10 years made money. For companies, it calculates debt-to-equity and interest cover and labels debt as Low, Moderate or High.

5.4 Machine Learning Module

This module (models.py) uses Spark MLlib and statsmodels. Features such as past returns, momentum over 3, 6 and 12 months, recent volatility and the market return are built for each of the 30 curated funds. Five models predict the next month's return: the historical mean (baseline), ARIMA(1,0,1), Linear Regression, Random Forest and Gradient-Boosted Trees. A second task predicts whether each fund will be in the calmest, middle or most volatile third of its peers in the next quarter, using Logistic Regression, Random Forest and a persistence baseline. All models are tested with walk-forward validation: they are trained on earlier years and tested on the next unseen year, from 2019 to 2026.

5.5 API Module

This module (main.py) is a FastAPI service that loads the Gold tables and serves them as JSON. It provides fifteen data endpoints, including /funds, /top-funds, /risk-scores, /holding-period, /models/comparison, /risk-prediction, /stocks and /indices, and a /health endpoint. A random 32-character API key is created once and stored in a file. The check_key function compares the X-API-Key header of every request with it and rejects a wrong key with error 401, so that only the website and Power BI can read the data.

5.6 Visualization and Dashboard Module

The website (index.html) asks for the API key and then shows what ₹1 lakh invested ten years ago is worth today, the top funds for any period, return against risk, holding-period rings, the model leaderboard, and the stocks and company

debt section. The Power BI dashboard connects to the same API using Power Query and has four pages: Home, Risk, Holding Period and Models, with slicers to filter by category or fund.

CHAPTER 6

IMPLEMENTATION

The project was implemented on a Windows laptop using WSL Ubuntu, with VS Code connected to WSL for editing and running the code. Java 11 and Python 3.12 were installed, and Hadoop 3.4.3 was configured in single-node mode with HDFS at hdfs://localhost:9000. A Python virtual environment holds PySpark, pandas, pyarrow, statsmodels, FastAPI and the other libraries.

The pipeline runs with a single command, run_pipeline.sh. It downloads the AMFI data, copies the Bronze files into HDFS under /project/bronze, and then runs three Spark jobs in order with spark-submit: silver.py for cleaning, gold.py for returns, risk and holding periods, and models.py for machine learning. The Gold tables are then exported for the API. A second command, run_stocks.sh, downloads the Nifty 50 stock, index and company debt data, loads it into HDFS and runs stocks.py. Each Spark job prints row counts and key results, and saves its log so that any failure can be traced.

The API is started with start_api.sh, which creates the API key on the first run and launches the FastAPI service with Uvicorn on port 8000. The website is served by the same service at http://localhost:8000. In Power BI Desktop, each table is loaded through a Blank Query that calls the API with the key in the X-API-Key header, after which the four dashboard pages were built. The full pipeline finishes in about five to ten minutes on a laptop.

CHAPTER 7

RESULTS AND DISCUSSION

The pipeline processed 37.05 million NAV records for 38,279 schemes into 31.2 million clean rows, and produced risk scores for 1,137 funds and 49 Nifty 50 stocks. All figures below are as of 1 October 2026.



Figure 7.1 Website home page

7.1 Returns of the Curated Funds

Mid-cap funds gave the highest returns. The best fund, Invesco India Midcap, grew at 18.96% a year, turning ₹1 lakh into ₹5.68 lakh in ten years, against ₹2.86 lakh for the Nifty 50 index fund. Every category fell by more than a third at its worst point during the March 2020 crash.

Table 7.1 Top five funds by 10-year return

Fund	10-year CAGR
Invesco India Midcap	18.96%
Nippon India Growth Mid Cap	17.67%
Kotak Midcap	17.23%
Axis Midcap	17.15%
Parag Parikh Flexi Cap	16.85%

7.2 Risk Score

Among the 30 curated funds, 8 were rated Low risk, 13 Medium and 9 High. Parag Parikh Flexi Cap had the lowest risk score (2.0) while still being fifth by return, and ICICI Prudential Nifty Next 50 Index had the highest (90.7). Figure 7.2 shows return against volatility for each fund.

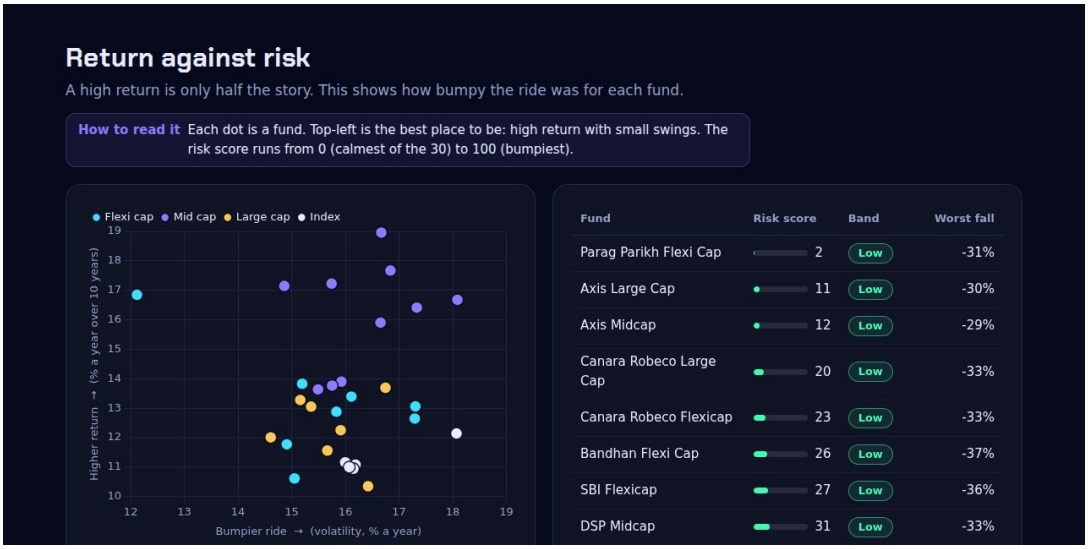


Figure 7.2 Return against risk

7.3 Holding Period

Holding longer clearly reduced the chance of loss. A 1-year holding made money in 83% of cases, while every holding of 7 years or more made money, with yearly returns between about 13% and 18%. These are past results from a mostly rising market and not a guarantee.

Table 7.2 Holding-period results (average of 30 funds)

Held for	Periods that made money	Median yearly return
1 year	83.0%	13.4%
3 years	97.8%	17.0%
5 years	99.9%	15.6%
7 years	100%	15.4%
10 years	100%	15.7%

7.4 Model Comparison

Out of 2,790 test predictions, only Random Forest beat the simple baseline, with 5.9% lower error and the correct direction 65.4% of the time. All models made their largest errors in 2020 because of the Covid crash. For the risk band of the next quarter, Logistic Regression was correct 61.2% of the time and Random Forest 60.1%, compared with 56.7% for the baseline and about 33% for a random guess.

Table 7.3 Forecasting model comparison

Model	RMSE	Direction right	vs baseline
Random Forest	0.0487	65.4%	5.9% better
Historical mean (baseline)	0.0518	64.1%	–
Linear Regression	0.0521	62.4%	0.7% worse
Gradient-Boosted Trees	0.0527	61.6%	1.7% worse
ARIMA(1,0,1)	0.0530	60.4%	2.3% worse

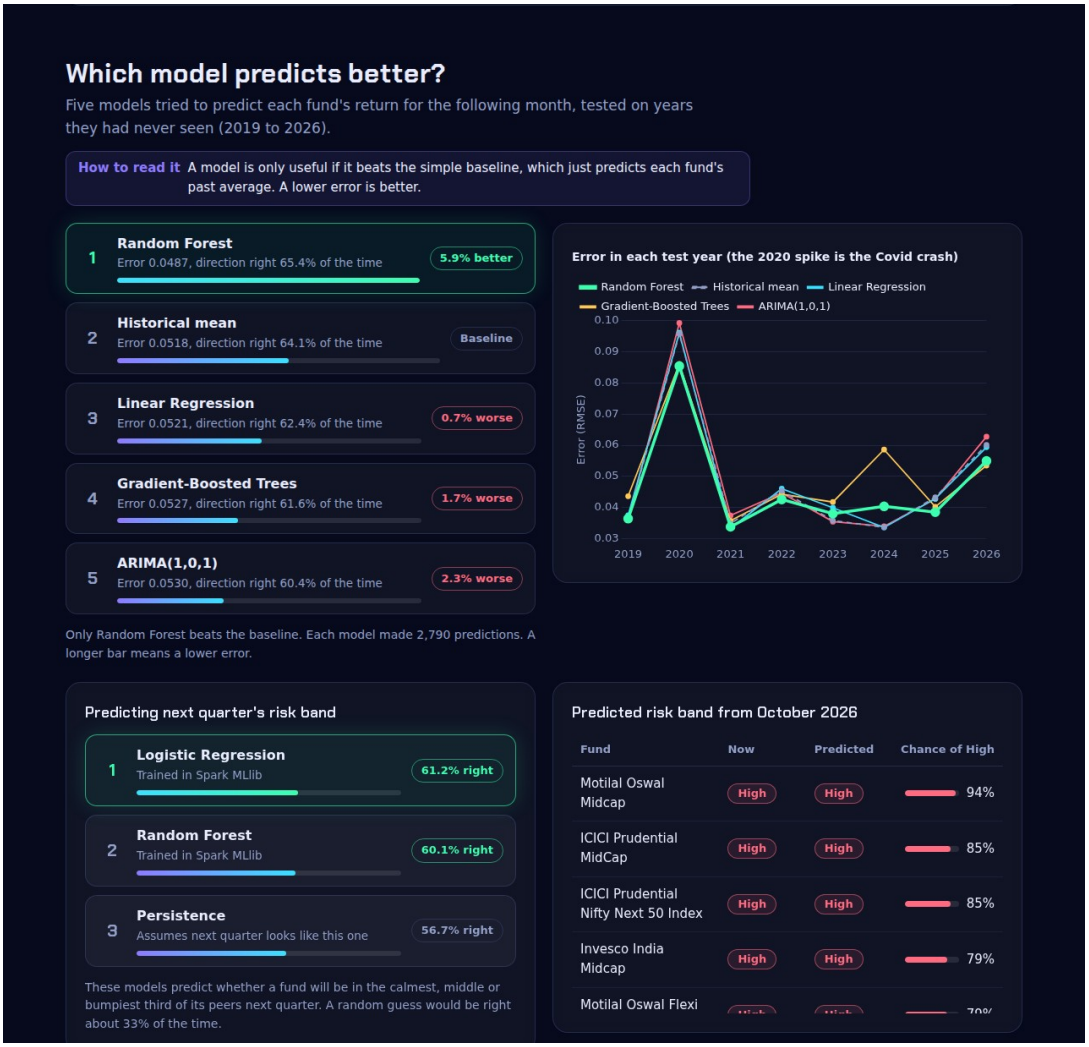


Figure 7.3 Model comparison

7.5 Stocks and Company Debt

Over ten years, ₹1 lakh in the Nifty 50 index grew to ₹2.60 lakh (10.0% a year) and in the Sensex to ₹2.58 lakh (9.9% a year). Of the 38 non-financial companies in the Nifty 50, 22 had low debt, 6 moderate and 10 high. InterGlobe Aviation had the highest debt-to-equity ratio at 11.15, with interest cover below 1. Banks and other financial companies were shown but not judged, because borrowing is their business.

Table 7.4 Company debt levels of Nifty 50 stocks

Debt level (debt-to-equity)	Number of companies
Low (up to 0.5)	22
Moderate (0.5 to 1)	6
High (above 1)	10
Financial companies (not judged)	12

CHAPTER 8

FUTURE ENHANCEMENTS

- **Multi-node cluster:** run Hadoop and Spark on several machines to handle more data and faster processing.
- **Scheduled updates:** run the pipeline automatically every day after new NAVs and prices are published.
- **Cloud hosting:** host the API online so that the Power BI dashboard can refresh on a schedule and be shared.
- **Index churn:** build a dated history of companies entering and leaving the Nifty 50 to study its effect on returns.
- **Longer debt history:** read company filings to extend debt analysis beyond four financial years.
- **Deep learning models:** test LSTM and hybrid models, and add news sentiment as an extra input.
- **Personal recommendations:** suggest funds that match an investor’s risk level and holding period.

CHAPTER 9

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